PAGE 1: BRANCH A — ADVANCED TECH & SYSTEMS

Masterclass Bible Mandate: UI/UX Architecture, Distributed Consensus, Threat Modeling & Web Graphics

WORKSPACE: Present-Career-os
BRANCH: main / agent/branch-a-tech

ASSIGNED WORKLOAD	COURSES COMPLETE	VOLUME TARGET	COLLISION STATUS
150 Days	**14 Courses**	**450,000 Wds**	**ZERO RISK**
5 Courses (Days 31–60)	DevOps, Cloud, Java, React...	≥ 600 words / part invariant	Isolated File Sovereignty

BRANCH A EXCLUSIVE COURSE ASSIGNMENT (150 REMAINING DAYS)

#	COURSE TITLE & ID	TARGET LONGLESSONS FILE	REMAINING	BIBLE SYLLABUS FOCUS (DAYS 31–60)
06	**UI/UX Design Systems** `course-design-systems`	`src/lib/data/designWebLongLessons.ts`	**30 Days (31–60)**	3-Tier Token Ramps, Component Variant Matrix, Web Components, Accessibility Tree, Radix UI & Design Engine.
13	**Full-Stack JavaScript** `course-fullstack-js`	`src/lib/data/fullstackJsLongLessons.ts`	**30 Days (31–60)**	Node.js C++ Addons, Event Loop Profiling, AsyncLocalStorage, Zero-Downtime Clustering, NestJS Enterprise Microservices.
15	**3D Interactive Graphics** `course-3d-graphics`	`src/lib/data/g3dLongLessons.ts`	**30 Days (31–60)**	WebGL/WebGPU Pipelines, GLSL Shader Compilation, Three.js BVH Spatial Partitioning, PBR Shading & VRoid Avatar Engine.
11	**Distributed Systems** `course-distributed-sys`	`src/lib/data/distributedWebLongLessons.ts`	**30 Days (31–60)**	Raft Consensus, Paxos Quorums, Distributed Sagas, Vector Clocks, CRDTs, Two-Phase Commit & Partition Healing.
09	**Cybersecurity Principles** `course-cybersecurity`	`src/lib/data/cyberWebLongLessons.ts`	**30 Days (31–60)**	STRIDE Threat Modeling, OAuth2/OIDC Grant Flows, mTLS Zero Trust, AST Taint Analysis, Cryptographic Key Escalation.

MONTH 2 BLOCK MILESTONES BLUEPRINT (BIBLE STANDARDS)

Course 06: Design Systems

- **Block 7 (31–35):** Semantic Tokens, CSS Ramps
- **Block 8 (36–40):** Headless Primitives, A11y
- **Block 9 (41–45):** Polymorphic Layout Engine
- **Block 10 (46–50):** Custom Web Components
- **Block 11 (51–55):** Micro-Frontend Theming
- **Block 12 (56–60):** Enterprise UI Capstone

Course 13: Full-Stack JS

- **Block 7 (31–35):** V8 Internals & Libuv Loop
- **Block 8 (36–40):** High-Throughput Streams
- **Block 9 (41–45):** Fastify/NestJS Microservices
- **Block 10 (46–50):** Distributed Redis Caching
- **Block 11 (51–55):** Memory Leak Flamegraphs
- **Block 12 (56–60):** Trading API Gateway

Course 15: 3D Graphics

- **Block 7 (31–35):** WebGL Buffers & GLSL Math
- **Block 8 (36–40):** Camera Frustum & Shadows
- **Block 9 (41–45):** Instanced Mesh Rendering
- **Block 10 (46–50):** WebGPU Compute Shaders
- **Block 11 (51–55):** VRoid Skeletal Rigs
- **Block 12 (56–60):** Campus 3D Metaverse

Course 11: Distributed Sys

- **Block 7 (31–35):** CAP/PACELC Trade-Offs
- **Block 8 (36–40):** Raft Leader Election & Log
- **Block 9 (41–45):** Eventual Consistency & CRDT
- **Block 10 (46–50):** Distributed Lock Manager
- **Block 11 (51–55):** Chaos Testing & Net Splits
- **Block 12 (56–60):** Resilient Ledger Capstone

Course 09: Cybersecurity

- **Block 7 (31–35):** Cryptographic Primitives
- **Block 8 (36–40):** Identity, JWT & PKCE
- **Block 9 (41–45):** SQLi, XSS & CSRF Exploits
- **Block 10 (46–50):** SAST/DAST Invariant Scans
- **Block 11 (51–55):** Zero Trust & Micro-Perimeter
- **Block 12 (56–60):** SOC2 Bank Audit Simulator

PEDAGOGICAL INVARIANTS FOR BRANCH A (BIBLE V8.0)

- **Slow-Reader Invariant:** Every single part MUST have ≥ 600 words in `say` array (3,000+ words/day).
- **Physical Analogies:** Zero unexplained jargon. Compare distributed consensus to physical jury voting.
- **Socratic Assessments:** 5 checks/day, 4 options, `answerIndex: 0`, statutory explanations ≥ 50 words.
- **Gate v2 Visuals:** `src/lib/data/lessonVisuals/{course}/day-{N}.json` (4 steps, SHA-256 bound, 0 digits).

NEGATIVE MANDATE & COLLISION FIREWALL

- **STRICTLY FORBIDDEN:** DO NOT touch `dsaWebLongLessons.ts` (Owned 100% by Branch C).
- **STRICTLY FORBIDDEN:** DO NOT touch any `bcom*` files (Owned 100% by Branch B).
- **Branch Isolation:** Work only inside `Present-Career-os` on assigned 5 courses.
- **Verification:** Run `npx tsc --noEmit` and test suite before every commit.

PAGE 2: BRANCH B — COMMERCE & DIGITAL BUSINESS

Masterclass Bible Mandate: Business Analytics, Predictive Modeling, Strategic Marketing & E-Commerce Systems

WORKSPACE: Present-Career-os-commerce
BRANCH: agent/branch-b-curriculum-commerce-iot

ASSIGNED WORKLOAD	COURSES COMPLETE	VOLUME TARGET	COLLISION STATUS
174 Days	**8 Courses**	**522,000 Wds**	**ZERO RISK**
5 Courses (25 M1 + 149 M2)	Accounting, Finance, Ops...	≥ 600 words / part invariant	Isolated File Sovereignty

BRANCH B EXCLUSIVE COURSE ASSIGNMENT (174 REMAINING DAYS)

#	COURSE TITLE & ID	TARGET LONGLESSONS FILE	REMAINING	BIBLE SYLLABUS FOCUS (DAYS 31–60)
24	**Business Analytics** `course-business-analytics`	`src/lib/data/bcomAnaLongLessons.ts`	**55 Days (6–60)**	Finish Month 1 Days 6–30 (25d) + Month 2 Days 31–60 (30d): Regression, Time Series, Monte Carlo, PowerBI & SQL Decision Engine.
25	**Marketing & Brand Mgmt** `course-marketing-branding`	`src/lib/data/bcomMktLongLessons.ts`	**30 Days (31–60)**	Brand Positioning, STP Framework, Customer Lifetime Value (CLV), Media Mix Modeling (MMM), Pricing Elasticity & Brand Equity.
26	**Digital Marketing Strategy** `course-digital-marketing`	`src/lib/data/bcomDmktLongLessons.ts`	**30 Days (31–60)**	Multi-Touch Attribution, CAC/LTV Unit Economics, SEO Technical Architecture, Programmatic Bidding, Conversion Optimization.
27	**E-Commerce & Digital Biz** `course-ecommerce-digital-biz`	`src/lib/data/bcomEcomLongLessons.ts`	**29 Days (32–60)**	Headless Shopify/Medusa, Checkout Microservices, Global Payments (PCI-DSS), Inventory Synchronization & Multi-Vendor Marketplaces.
29	**Sales, CRM & Success** `course-sales-crm-success`	`src/lib/data/bcomScrmLongLessons.ts`	**30 Days (31–60)**	MEDDIC & Challenger Sales Methodology, Salesforce CRM Architecture, Churn Prediction, Pipeline Forecasting & Account-Based Selling.

MONTH 2 BLOCK MILESTONES BLUEPRINT (BIBLE STANDARDS)

Course 24: Analytics

M1 Days 6–30 (25d): SQL, Stats, EDA
Block 7 (31–35): Multivariate Regression
Block 8 (36–40): ARIMA Forecasting
Block 9 (41–45): Cohort Churn Modeling
Block 10 (46–50): BigQuery Optimization
Block 12 (56–60): Executive BI Capstone

Course 25: Brand Mgmt

Block 7 (31–35): Brand Architecture
Block 8 (36–40): Perceptual Mapping
Block 9 (41–45): Pricing & Conjoint Analysis
Block 10 (46–50): Crisis Brand PR Defense
Block 11 (51–55): Global Market Expansion
Block 12 (56–60): Brand Audit & Valuation

Course 26: Digital Mkt

Block 7 (31–35): Data-Driven Attribution
Block 8 (36–40): Performance Funnels
Block 9 (41–45): Paid Search RTB Algorithms
Block 10 (46–50): Retention CRM Automation
Block 11 (51–55): Cookie-Less Tracking APIs
Block 12 (56–60): $10M Ad Budget Strategy

Course 27: E-Commerce

Block 7 (32–35): Headless Composable UI
Block 8 (36–40): Cart & State Synchronization
Block 9 (41–45): Stripe/Adyen Payment Rails
Block 10 (46–50): Warehouse OMS Real-Time
Block 11 (51–55): Fraud Detection Systems
Block 12 (56–60): Global Marketplace Launch

Course 29: Sales & CRM

Block 7 (31–35): MEDDIC Qualification
Block 8 (36–40): Complex Enterprise RFPs
Block 9 (41–45): CRM Entity Relationships
Block 10 (46–50): CS Health Scoring Engine
Block 11 (51–55): Contract Redlining Mastery
Block 12 (56–60): $5M Deal Close Simulator

PEDAGOGICAL INVARIANTS FOR BRANCH B (BIBLE V8.0)

- **Slow-Reader Invariant:** Every single part MUST have ≥ 600 words in `say` array (3,000+ words/day).
- **Executive Rigor:** Real financial spreadsheets, SQL schemas, and marketing math. No abstract filler.
- **Socratic Assessments:** 5 checks/day, 4 options, `answerIndex: 0`, comprehensive statutory explanations.
- **Gate v2 Visuals:** `src/lib/data/lessonVisuals/{course}/day-{N}.json` (100% verified schema).

NEGATIVE MANDATE & COLLISION FIREWALL

- **STRICTLY FORBIDDEN:** DO NOT touch Tech courses (`java`, `devops`, `dsa`, `react`, `node`).
- **STRICTLY FORBIDDEN:** DO NOT touch AI courses assigned to Branch C (`prompt`, `nlp`, `ai-eng`).
- **Branch Isolation:** Work only inside `Present-Career-os-commerce` on assigned Commerce files.
- **Verification:** Push block-by-block and update central `PROJECT_MEMORY.md` upon completion.

PAGE 3: BRANCH C — ALGORITHMS & APPLIED AI

Masterclass Bible Mandate: Algorithmic Optimization, Machine Learning Systems, NLP & Enterprise Automation

WORKSPACE: Present-Career-os-platform
BRANCH: agent/branch-c-platform-admin-faculty

ASSIGNED WORKLOAD	ACTIVE IN-FLIGHT	VOLUME TARGET	COLLISION STATUS
143 Days	**DSA Day 38**	**429,000 Wds**	**ZERO RISK**
5 Courses (Month 2 Days 31–60)	Days 1–37 Complete & Pushed	≥ 600 words / part invariant	Isolated File Sovereignty

BRANCH C EXCLUSIVE COURSE ASSIGNMENT (143 REMAINING DAYS)

#	COURSE TITLE & ID	TARGET LONGLESSONS FILE	REMAINING	BIBLE SYLLABUS FOCUS (DAYS 31–60)
07	**Algorithms & Optimizations** `course-dsa-optim`	`src/lib/data/dsaWebLongLessons.ts`	**23 Days (38–60)**	**IN PROGRESS (Day 38 active)**: KMP, Aho-Corasick, Suffix Automata, Ford-Fulkerson Max-Flow, Simplex & High-Perf Memory Profiling.
12	**Applied AI & LLM Systems** `course-ai-eng`	`src/lib/data/aiWebLongLessons.ts`	**30 Days (31–60)**	Vector Databases, HNSW Indexing, Multi-Agent RAG Orchestration, Context Window Management, LoRA Fine-Tuning & Quantization.
33	**AI Literacy & Prompt Eng** `course-ai-prompt-literacy`	`src/lib/data/promptPythonLongLessons.ts`	**30 Days (31–60)**	Chain-of-Thought, ReAct Prompting, Constitutional AI, Structured JSON Output, Prompt Injection Defense & Evaluation Harnesses.
37	**NLP & Speech Tech** `course-nlp`	`src/lib/data/nlpPythonLongLessons.ts`	**30 Days (31–60)**	Tokenization Algorithms (BPE/WordPiece), Transformer Self-Attention Math, BERT Encoders, Whisper ASR, TTS Speech Synthesis.
31	**AI Digital Transformation** `course-ai-digital-transformation`	`src/lib/data/bcomAitLongLessons.ts`	**30 Days (31–60)**	Enterprise Automation Architecture, LangGraph Agent Workflows, Robotic Process Automation (RPA), FinTech AI Governance & ROI Auditing.

MONTH 2 BLOCK MILESTONES BLUEPRINT (BIBLE STANDARDS)

Course 07: DSA Optim

- **Block 8 (36–40):** String Automata (KMP)
- **Block 9 (41–45):** Network Flow & Cut
- **Block 10 (46–50):** Segment Trees & Fenwick
- **Block 11 (51–55):** Cache-Oblivious Structs
- **Block 12 (56–60):** HFT Router Capstone

Course 12: AI Engineering

- **Block 7 (31–35):** Embeddings & Cosine Space
- **Block 8 (36–40):** Advanced Chunking & RAG
- **Block 9 (41–45):** Multi-Agent LangGraph
- **Block 10 (46–50):** Model Quantization (GGUF)
- **Block 11 (51–55):** Tool Calling & Sandboxes
- **Block 12 (56–60):** Autonomous AI SRE Agent

Course 33: Prompt Literacy

- **Block 7 (31–35):** Few-Shot In-Context Rules
- **Block 8 (36–40):** Structured JSON Schemas
- **Block 9 (41–45):** Guardrails & Red-Teaming
- **Block 10 (46–50):** Context Compaction & Cache
- **Block 11 (51–55):** Automated Prompt Evals
- **Block 12 (56–60):** Enterprise Prompt System

Course 37: NLP & Speech

- **Block 7 (31–35):** Subword Tokenizer Math
- **Block 8 (36–40):** Positional Encodings
- **Block 9 (41–45):** Encoder-Decoder Seq2Seq
- **Block 10 (46–50):** Whisper Audio Spectrograms
- **Block 11 (51–55):** Real-Time Voice Streaming
- **Block 12 (56–60):** Voice Agent Copilot

Course 31: AI Transformation

- **Block 7 (31–35):** Legacy System Assessment
- **Block 8 (36–40):** LLM Gateway Architecture
- **Block 9 (41–45):** Data Privacy & On-Prem AI
- **Block 10 (46–50):** Autonomous Workflow Bots
- **Block 11 (51–55):** AI FinOps & Token Costs
- **Block 12 (56–60):** Boardroom AI Roadmap

PEDAGOGICAL INVARIANTS FOR BRANCH C (BIBLE V8.0)

- **Slow-Reader Invariant:** Every single part MUST have ≥ 600 words in `say` array (3,000+ words/day).
- **Executable Code:** All algorithms, vector math, and Python/TypeScript snippets must run deterministically.
- **Socratic Assessments:** 5 checks/day, 4 options, `answerIndex: 0`, deep mathematical/algorithmic whys.
- **Gate v2 Visuals:** `src/lib/data/lessonVisuals/{course}/day-{N}.json` (4 synchronized visual steps).

NEGATIVE MANDATE & COLLISION FIREWALL

- **STRICTLY FORBIDDEN:** DO NOT touch Branch A courses (`design-systems`, `cybersecurity`, `distributed`, etc.).
- **STRICTLY FORBIDDEN:** DO NOT touch Branch B courses (`bcom_ana`, `bcom_mkt`, `bcom_dmkt`, `bcom_ecom`).
- **Finish DSA First:** Complete Days 38–60 of `dsaWebLongLessons.ts` before picking up Course 12.
- **Verification:** Run `npx tsc --noEmit` and test suite before every commit.